The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

(Dated: March 12, 2026)

Fock's 1935 stereographic projection maps the hydrogen atom onto the unit three-sphere S^3, where the Schrödinger equation becomes a pure Laplace–Beltrami eigenvalue problem with no potential term. The GeoVac program discretizes this S^3 as a graph Laplacian, achieving $< 0.1\%$ error for one-electron atoms with zero free parameters (Papers 0, 1, 7). For molecules, however, S^3 is not the natural geometry: the single-p_0 momentum scale cannot encode the R-dependent bonding physics (Paper 8–9, negative theorem).

We resolve this by identifying the *natural geometry principle*: every separable quantum system has a natural lattice—the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where separation occurs. For one-center systems that geometry is S^3; for two-center systems it is the prolate spheroid. We discretize the separated prolate spheroidal equations for H_2^+ using a spectral angular solver (Legendre basis) and a self-adjoint finite-difference radial solver, with root-finding in the separation parameter $c^2 = -R^2 E_{\text{elec}}/2$ to enforce self-consistency. The method has *zero* free parameters: all inputs are nuclear charges, internuclear distance, and grid resolution.

Results for H_2^+: equilibrium bond length $R_{\text{eq}} = 2.001$ bohr (0.21% error vs. exact 1.997 bohr), minimum energy $E_{\min} = -0.5984$ Ha (0.70% error vs. exact -0.6026 Ha), and correct dissociation to H + p at $R \to \infty$. Grid convergence is $O(1/N_\xi)$ in the radial grid, with sub-percent accuracy at $N_\xi = 5000$. This establishes the prolate spheroidal lattice as the molecular analog of Fock's atomic S^3 projection.

I. INTRODUCTION

The Geometric Vacuum program rests on a simple observation: the hydrogen atom's Schrödinger equation, when projected onto momentum space via Fock's 1935 conformal map [8], becomes the eigenvalue problem for the Laplace–Beltrami operator on the unit three-sphere S^3. The Coulomb potential $-Z/r$ is not an input force law but a coordinate distortion produced by the stereographic projection from S^3 to flat $\mathbb{R}^3$. This equivalence, established symbolically with 18 independent proofs [3], underlies the entire GeoVac computational framework: discretize S^3 as a graph, solve the graph Laplacian eigenvalue problem, and recover atomic energies to $< 0.1\%$ error with no fitted parameters [1, 2].

The extension to molecules has proven more subtle. Paper 8–9 established the *bond sphere* picture—a diatomic molecule as a single S^3 with two weighted poles—and derived the $SO(4)$ Wigner D-matrix elements that rotate atomic orbitals from one center to another [4]. However, that same paper proved a negative structural theorem: in the molecular Sturmian basis with a shared momentum scale p_0, the Hamiltonian is proportional to the overlap matrix ($H_{ij} \propto S_{ij}$), rendering the generalized eigenvalues independent of R. Binding in this framework requires the full R-dependent eigenvalue spectrum $\beta_k(R)$ from the continuous prolate spheroidal PDE—reintroducing the very continuous spatial geometry the graph was meant to replace.

For heteronuclear systems, the situation is worse: no shared p_0 exists that simultaneously represents both atomic length scales. The LCAO approach (atom-centered lattices with bridges) partially circumvents this, achieving $D_e^{\text{CP}} = 0.093$ Ha for LiH at nmax= 3 (1.0% error) [5], but carries intrinsic BSSE and an R-independent kinetic energy that shifts R_{eq} to ~ 2.5 bohr (expt. 3.015).

This paper takes a fundamentally different approach. Rather than forcing molecules onto S^3, we ask: *what is the natural geometry for two-center systems?* The answer is immediate from classical mathematical physics: the two-center Coulomb problem separates in prolate spheroidal coordinates (ξ, η, φ), just as the one-center problem separates in spherical coordinates [9, 10]. Fock's projection exploits spherical separation via $SO(4)$ symmetry; the molecular analog exploits prolate spheroidal separation.

We formalize this as the **natural geometry principle**:

> *Every separable quantum system has a natural lattice—the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where separation occurs.*

For atoms, the natural lattice is the S^3 graph. For diatomics, it is the prolate spheroidal lattice. The GeoVac program is not about S^3 specifically—it is about identifying and discretizing the natural conformal geometry for each class of quantum systems.

The remainder of this paper implements this principle for H_2^+, the simplest diatomic. Section II reviews prolate spheroidal coordinates. Section III derives the separated equations. Sections IV and V describe the angular (spec-

* Paper 11 in the Geometric Vacuum series

tral) and radial (finite-difference) solvers. Section VI explains the self-consistency root-finding. Section VII presents H_2^+ results: PES, spectroscopic constants, and grid convergence. Section VIII discusses implications for the GeoVac program. Section IX concludes.

II. PROLATE SPHEROIDAL COORDINATES

Place two nuclei A and B at the foci of a prolate spheroid, separated by distance R. The prolate spheroidal coordinates (ξ, η, φ) are defined by

$$\xi = \frac{r_A + r_B}{R}, \qquad \xi \in [1, \infty), \tag{1}$$

$$\eta = \frac{r_A - r_B}{R}, \qquad \eta \in [-1, +1], \tag{2}$$

$$\varphi = \arctan(y/x), \qquad \varphi \in [0, 2\pi), \tag{3}$$

where r_A and r_B are distances from the electron to nuclei A and B. Surfaces of constant ξ are confocal ellipsoids (with $\xi = 1$ the degenerate line segment joining the foci); surfaces of constant η are confocal hyperboloids ($\eta = \pm 1$ are the two half-axes beyond the nuclei).

The metric tensor has diagonal elements

$$g_{\xi\xi} = \frac{R^2}{4}\frac{\xi^2 - \eta^2}{\xi^2 - 1}, \quad g_{\eta\eta} = \frac{R^2}{4}\frac{\xi^2 - \eta^2}{1 - \eta^2}, \quad g_{\varphi\varphi} = \frac{R^2}{4}(\xi^2 - 1)(1 - \eta^2). \tag{4}$$

and the volume element is

$$dV = \frac{R^3}{8}(\xi^2 - \eta^2)\, d\xi\, d\eta\, d\varphi. \tag{5}$$

The nuclear distances take the simple forms

$$r_A = \frac{R}{2}(\xi + \eta), \qquad r_B = \frac{R}{2}(\xi - \eta), \tag{6}$$

so the nuclear attraction potential for charges Z_A and Z_B is

$$V = -\frac{Z_A}{r_A} - \frac{Z_B}{r_B} = -\frac{2}{R}\left[\frac{(Z_A + Z_B)\xi + (Z_B - Z_A)\eta}{\xi^2 - \eta^2}\right]. \tag{7}$$

The Laplacian in prolate spheroidal coordinates is

$$\nabla^2 = \frac{4}{R^2(\xi^2 - \eta^2)}\left[\frac{\partial}{\partial\xi}\left((\xi^2 - 1)\frac{\partial}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left((1 - \eta^2)\frac{\partial}{\partial\eta}\right) + \frac{\xi^2 - \eta^2}{(\xi^2 - 1)(1 - \eta^2)}\frac{\partial^2}{\partial\varphi^2}\right]. \tag{8}$$

III. SEPARATION OF THE TWO-CENTER COULOMB PROBLEM

The one-electron Schrödinger equation $(-\frac{1}{2}\nabla^2 + V)\Psi = E_{\text{elec}}\Psi$ with the potential (7) can be written, after multiplying both sides by $-R^2(\xi^2 - \eta^2)/4$, as

$$\left[\hat{L}_\xi + \hat{L}_\eta + \frac{\xi^2 - \eta^2}{(\xi^2 - 1)(1 - \eta^2)}\frac{\partial^2}{\partial\varphi^2}\right]\Psi = 0, \tag{9}$$

where we define the ξ- and η-operators

$$\hat{L}_\xi = \frac{\partial}{\partial\xi}\left[(\xi^2 - 1)\frac{\partial}{\partial\xi}\right] + a\xi - c^2\xi^2, \tag{10}$$

$$\hat{L}_\eta = \frac{\partial}{\partial\eta}\left[(1 - \eta^2)\frac{\partial}{\partial\eta}\right] + b\eta + c^2\eta^2, \tag{11}$$

with the dimensionless parameters

$$a = R(Z_A + Z_B), \quad b = R(Z_B - Z_A), \quad c^2 = -\frac{R^2 E_{\text{elec}}}{2}. \tag{12}$$

For bound states ($E_{\text{elec}} < 0$), $c^2 > 0$.

The ansatz $\Psi(\xi, \eta, \varphi) = F(\xi)\, G(\eta)\, e^{im\varphi}$ separates Eq. (9) into two ordinary differential equations coupled through a separation constant A:

a. ξ-equation (radial):

$$\frac{d}{d\xi}\left[(\xi^2 - 1)\frac{dF}{d\xi}\right] + \left(A + a\xi - c^2\xi^2 - \frac{m^2}{\xi^2 - 1}\right)F = 0. \tag{13}$$

b. η-equation (angular):

$$\frac{d}{d\eta}\left[(1 - \eta^2)\frac{dG}{d\eta}\right] + \left(-A + c^2\eta^2 + b\eta - \frac{m^2}{1 - \eta^2}\right)G = 0. \tag{14}$$

The angular equation (14) is a generalization of the associated Legendre equation with additional $c^2\eta^2$ and $b\eta$ terms. For homonuclear diatomics ($Z_A = Z_B$), $b = 0$ and the η-equation reduces to the spheroidal wave equation. The radial equation (13) has an irregular singular point at $\xi = 1$ (the line segment between the nuclei) and an irregular singular point at $\xi \to \infty$.

The key feature of the separated system is that both equations (13) and (14) contain the same three unknowns: the energy E_{elec} (via c^2), the separation constant A, and the quantum numbers (m, n_ξ, n_η). For fixed m and state indices, the problem reduces to finding the value of c^2 for which both equations yield the same A.

IV. THE ANGULAR EQUATION: SPECTRAL SOLVER

The angular equation (14) is an eigenvalue problem for A at fixed c^2. We solve it using a spectral method with Legendre polynomial basis functions [11].

A. Legendre expansion

Expand $G(\eta) = \sum_{l=|m|}^{l_{\max}} a_l P_l^{|m|}(\eta)$ in associated Legendre polynomials. Substituting into Eq. (14) and using the recurrence relations for P_l^m produces a symmetric tridiagonal (pentadiagonal for $b \neq 0$) generalized eigenvalue problem

$$\mathbf{M}\,\mathbf{a} = A\,\mathbf{a}, \tag{15}$$

where the matrix $\mathbf{M}$ has elements determined by the three-term recurrence for $\eta^2 P_l^m$ and (when $b \neq 0$) ηP_l^m.

For homonuclear diatomics ($b = 0$), the matrix $\mathbf{M}$ is symmetric and tridiagonal, and the eigenvalues converge exponentially in $l_{\max}$. With $l_{\max} = 50$, the ground-state separation constant A is converged to machine precision for all c^2 values of interest.

B. Boundary conditions

The associated Legendre polynomials automatically satisfy the correct boundary conditions: regularity at $\eta = \pm 1$. No additional boundary conditions need to be imposed, because the factor $(1-\eta^2) \to 0$ at the endpoints provides a natural zero-flux condition for the self-adjoint operator.

For homonuclear diatomics, the solutions have definite parity under $\eta \to -\eta$: even parity (σ_g, π_u, ...) or odd parity (σ_u, π_g, ...). This parity is automatically captured by the Legendre expansion through the selection of even or odd l values.

V. THE RADIAL EQUATION: SELF-ADJOINT FINITE DIFFERENCES

The radial equation (13) is solved numerically on a uniform grid in ξ. The critical design choice is the discretization of the self-adjoint differential operator $d/d\xi\,[p(\xi)\,dF/d\xi]$ where $p(\xi) = \xi^2 - 1$.

A. Self-adjoint stencil

The standard second-order self-adjoint stencil [14] for $d/d\xi\,[p(\xi)\,dF/d\xi]$ on a uniform grid with spacing h evaluates p at the half-points:

$$\frac{1}{h^2}\left[p_{i+1/2}(F_{i+1} - F_i) - p_{i-1/2}(F_i - F_{i-1})\right], \tag{16}$$

where $p_{i\pm 1/2} = (\xi_i \pm h/2)^2 - 1$. This yields a symmetric tridiagonal matrix with diagonal elements

$$d_i = -\frac{p_{i+1/2} + p_{i-1/2}}{h^2} + A + a\xi_i - c^2\xi_i^2 - \frac{m^2}{\xi_i^2 - 1} \tag{17}$$

and off-diagonal elements

$$e_i = \frac{p_{i+1/2}}{h^2}. \tag{18}$$

B. Boundary conditions

Two boundary conditions are required:

a. At $\xi = 1$ (nuclear axis): For σ states ($m = 0$), the wavefunction is nonzero on the internuclear axis. Since $p(1) = 0$, the self-adjoint operator has a natural *Neumann* boundary condition at $\xi = 1$: the flux $p(\xi)\,dF/d\xi \to 0$ as $\xi \to 1^+$ regardless of the value of $dF/d\xi$. Computationally, this is implemented by setting $p_{1/2} = 0$ in the stencil (16), which modifies the first diagonal element to $d_1 = -p_{3/2}/h^2 + q_1$.

For π states ($m \neq 0$), the wavefunction vanishes on the axis by the $e^{im\varphi}$ factor, and a Dirichlet condition $F(1) = 0$ is appropriate.

b. At $\xi = \xi_{\max}$ (asymptotic region): A Dirichlet condition $F(\xi_{\max}) = 0$ is imposed, with $\xi_{\max}$ chosen large enough that the bound-state wavefunction has decayed to negligible amplitude. For H_2^+ near equilibrium, $\xi_{\max} = 25$ is sufficient; for large R, $\xi_{\max}$ is scaled with R.

C. Eigenvalue interpretation

With A fixed from the angular solver, the radial operator (13) is a Sturm–Liouville problem parameterized by c^2. At the *correct* c^2, the largest eigenvalue of the discretized operator equals zero—the bound state sits exactly at the top of the spectrum. At other values of c^2, the top eigenvalue is nonzero. This provides a scalar residual function $f(c^2) = \lambda_{\max}(c^2)$ for root-finding (Section VI).

The grid offset $\xi_{\min} = 1 + \delta$ (with $\delta = 5 \times 10^{-4}$) avoids the singular point $\xi = 1$ while keeping the first grid point close enough to capture the near-nuclear wavefunction.

VI. ROOT-FINDING AND SELF-CONSISTENCY

The complete algorithm proceeds as follows:

1. **Input:** Z_A, Z_B, R, grid parameters (N_ξ, $\xi_{\max}$), azimuthal quantum number m.
2. **Define residual** $f(c^2)$:
   (a) Compute $A(c^2)$ from the angular spectral solver (Section IV).
   (b) Compute $\lambda_{\max}(c^2, A)$ from the radial FD solver (Section V).
   (c) Return $f = \lambda_{\max}$.
3. **Bracket:** Find $c^2_{\min}$ and $c^2_{\max}$ such that $f(c^2_{\min}) > 0$ and $f(c^2_{\max}) < 0$. For homonuclear H_2^+, typical brackets are $c^2 \in [0.01, R^2(Z_A + Z_B)^2]$.
4. **Root-find:** Apply Brent's method [13] to solve $f(c^2) = 0$ to tolerance 10^{-12}.
5. **Extract energy:** $E_{\text{elec}} = -2c^2/R^2$, $E_{\text{total}} = E_{\text{elec}} + Z_A Z_B/R$.

The algorithm is completely parameter-free: the only inputs are the physical quantities (Z_A, Z_B, R) and the numerical resolution (N_ξ, $\xi_{\max}$). No orbital exponents, basis set parameters, kinetic scaling factors, or bridge coupling constants appear anywhere.

Each evaluation of $f(c^2)$ requires one angular eigenvalue solve ($O(l_{\max})$ for tridiagonal) and one radial eigenvalue solve ($O(N_\xi)$ for tridiagonal). Brent's method typically converges in ~ 50 iterations, so the total cost per energy point is $O(50 \times N_\xi) \sim O(N_\xi)$. For $N_\xi = 5000$, a single point takes approximately 0.2 s on commodity hardware.

VII. RESULTS: H_2^+ GROUND STATE

A. Grid convergence

Table I shows the convergence of the ground-state total energy for H_2^+ at $R = 2.0$ bohr as a function of radial grid size N_ξ (with $\xi_{\max} = 25$). The exact value [9] is $E_{\text{total}} = -0.6026$ Ha.

TABLE I. Grid convergence for H_2^+ at $R = 2.0$ bohr. The exact total energy is -0.6026 Ha.

N_ξ	c^2	E_{total} (Ha)	Error (%)
500	2.114	−0.5568	7.60
1000	2.156	−0.5779	4.11
2000	2.179	−0.5893	2.21
5000	2.193	−0.5965	1.01
10000	2.198	−0.5990	0.59

The convergence is approximately $O(1/N_\xi)$, consistent with second-order finite-difference discretization of the radial equation. Sub-percent accuracy is achieved at $N_\xi = 5000$ (0.2 s wall time). Higher-order FD stencils or spectral radial methods would improve this convergence rate.

B. Potential energy surface

Figure **??** shows the H_2^+ potential energy surface computed with $N_\xi = 8000$ over $R \in [1.0, 6.0]$ bohr. The curve exhibits a clear minimum near $R = 2.0$ bohr and dissociates correctly toward $E = -0.5$ Ha (the H atom ground state plus a bare proton) at large R.

C. Spectroscopic constants

Table III compares the computed spectroscopic constants with exact values. Equilibrium geometry and binding energy are obtained from a quadratic fit to the PES near the minimum.

TABLE II. H_2^+ potential energy surface near the minimum ($N_\xi = 8000$, $\xi_{\max} = 25$).

R (bohr)	E_{total} (Ha)
1.50	−0.5783
1.60	−0.5869
1.70	−0.5926
1.80	−0.5961
1.90	−0.5979
2.00	−0.5984
2.10	−0.5979
2.20	−0.5965
2.50	−0.5894
3.00	−0.5730
4.00	−0.5412
5.00	−0.5190
6.00	−0.5058

TABLE III. Spectroscopic constants for H_2^+ ($N_\xi = 8000$). Exact values from Bates *et al.* [9].

Quantity	Computed	Exact	Error (%)
R_{eq} (bohr)	2.001	1.997	0.21
$E_{\min}$ (Ha)	−0.5984	−0.6026	0.70
D_e (Ha)	0.0984	0.1026	4.09
k (Ha/bohr^2)	0.1032	~ 0.10	~ 3

The equilibrium bond length is reproduced to 0.21% and the minimum energy to 0.70%. The dissociation energy $D_e = -0.5 - E_{\min}$ has a larger relative error (4.09%) because it is the small difference between two large numbers. The force constant k agrees to $\sim 3\%$.

D. Dissociation limit

At large R, the electronic energy approaches $E_{\text{elec}} \to -1/2$ (the hydrogen atom ground state) and the total energy approaches $E_{\text{total}} \to -1/2$ (since $V_{NN} = 1/R \to 0$). At $R = 10$ bohr with increased grid parameters ($N_\xi = 8000$, $\xi_{\max} = 80$), we obtain $E_{\text{total}} = -0.476$ Ha, approaching the exact limit of -0.500 Ha. The slow convergence at large R reflects the need for a large $\xi_{\max}$ to capture the diffuse wavefunction.

E. Heteronuclear extension: HeH^{2+}

The method extends naturally to heteronuclear one-electron systems. For HeH^{2+} ($Z_A = 2$, $Z_B = 1$) at $R = 2.0$ bohr, we obtain $E_{\text{total}} = -1.475$ Ha. The heteronuclear case introduces a nonzero $b = R(Z_B - Z_A)$ parameter in the angular equation, breaking the $\eta \to -\eta$ parity symmetry, but the algorithm handles this without modification.

VIII. DISCUSSION

A. The natural geometry principle

The results of this paper, combined with the atomic results of Papers 0, 1, and 7, establish a pattern:

- **One-center systems (atoms):** The natural geometry is the three-sphere S^3, accessed via Fock's stereographic projection from momentum space. The Schrödinger equation becomes the Laplace–Beltrami eigenvalue problem on S^3, with no potential term.
- **Two-center systems (diatomics):** The natural geometry is the prolate spheroid, where the two-center Coulomb problem separates into two coupled ODEs. The bond length R enters through the metric and the separation parameter c^2.

In both cases, the strategy is identical: identify the coordinate system where the Schrödinger equation separates, discretize the resulting Laplace–Beltrami operator, and solve the resulting eigenvalue problem. The physics—nuclear charges, distances, masses—enters only through the coordinate system and its metric. There are no basis set parameters, no fitted constants, and no orbital exponents to optimize.

B. Resolution of the bond sphere impasse

Paper 8–9 proved that the molecular Sturmian basis on a single S^3 with shared momentum scale p_0 cannot produce R-dependent binding: the Hamiltonian is proportional to the overlap matrix, making eigenvalues geometry-independent. The prolate spheroidal lattice resolves this impasse by abandoning the single-center S^3 geometry entirely. Rather than forcing two centers onto one sphere, each center retains its identity within the natural two-center geometry.

The bond length R does not enter as an external parameter grafted onto an R-independent topology. Instead, R is intrinsic to the prolate spheroidal metric (4): it sets the interfocal distance, which determines the coordinate system itself. This is analogous to how p_0 is intrinsic to the Fock projection for atoms—it is not a free parameter but is determined self-consistently by the energy eigenvalue.

C. Comparison with LCAO approach

The LCAO approach [5] constructs a molecular basis by combining atom-centered lattices with bridge elements. While successful for energetics ($D_e^{\mathrm{CP}} = 0.093$ Ha for LiH, 1.0% error), it has inherent limitations:

1. **BSSE:** Combining two lattices creates basis set superposition error (-0.115 Ha for LiH at nmax= 3), requiring counterpoise correction.
2. **R-independent kinetic energy:** The graph Laplacian eigenvalues on each atomic lattice do not depend on R, shifting R_{eq} to ~ 2.5 bohr (expt. 3.015).
3. **Bridge parameters:** The inter-atomic coupling requires STO overlap integrals and Mulliken-type approximations.

The prolate spheroidal lattice has none of these limitations: no BSSE (single coordinate system), R-dependent metric (correct R_{eq}), and no fitted parameters. For one-electron systems, it is clearly superior.

D. Limitations and extensions

The current implementation is limited to one-electron diatomics (H_2^+, HeH^{2+}, etc.), where the Schrödinger equation separates exactly. The most important extensions are:

a. Two-electron systems (H_2). With two electrons, the Schrödinger equation no longer separates in prolate spheroidal coordinates due to the electron–electron repulsion $1/r_{12}$. A prolate spheroidal CI approach would construct single-particle orbitals from the separated H_2^+ solutions and build Slater determinants for the two-electron problem. The V_{ee} matrix elements in prolate spheroidal coordinates are analytically tractable [12].

b. Excited states. The current solver targets the ground σ_g state. Excited states (σ_u, π_g, π_u) are accessible by changing the azimuthal quantum number m and selecting different eigenvalues of the angular and radial operators.

c. Higher-order discretization. The $O(1/N_\xi)$ convergence of the second-order FD scheme could be improved with fourth-order stencils or spectral (Chebyshev or mapped Legendre) methods for the radial equation, potentially achieving sub-0.1% accuracy with $N_\xi \sim 500$.

IX. CONCLUSION

We have demonstrated that the natural geometry for diatomic molecules is the prolate spheroid, just as the natural geometry for atoms is the three-sphere S^3. By discretizing the separated prolate spheroidal equations with a spectral angular solver and a self-adjoint finite-difference radial solver, we obtain the H_2^+ ground-state potential energy surface to 0.70% energy accuracy and 0.21% bond length accuracy with zero free parameters.

This result establishes the *natural geometry principle* as the organizing idea of the GeoVac program:

> *Every separable quantum system has a natural lattice—the lattice that discretizes the*

Laplace–Beltrami operator in the coordinate system where separation occurs.

The atomic S^3 lattice (Papers 0, 1, 7) and the molecular prolate spheroidal lattice (this paper) are the first two instances of this principle. Future work will extend the approach to two-electron diatomics (H_2), polyatomic systems with other natural coordinate systems, and higher-order discretization methods for improved convergence.

The prolate spheroidal lattice also resolves the impasse identified in Paper 8–9: the single-p_0 Sturmian basis on S^3 cannot produce R-dependent binding, but the prolate spheroidal coordinate system encodes R directly in its metric, producing the correct equilibrium geometry without any fitted parameters.

ACKNOWLEDGMENTS

Computational resources were provided by personal hardware.

Appendix A: Grid Convergence Analysis

The second-order self-adjoint finite-difference stencil gives truncation error $O(h^2) = O((\xi_{\max}/N_\xi)^2)$. Since the top eigenvalue of the discretized radial operator must equal zero, the energy error is controlled by the eigenvalue error, which scales as $O(h^2)$ for the largest eigenvalue.

In practice, the observed convergence rate is closer to $O(1/N_\xi)$ (Table I) rather than $O(1/N_\xi^2)$. This suboptimal convergence is attributed to:

1. The grid offset $\delta = 5\times10^{-4}$ from the singular point $\xi = 1$, which introduces an $O(\delta)$ boundary error.
2. The Neumann boundary condition implementation, which is first-order at the boundary.
3. The nonuniform behavior of $p(\xi) = \xi^2 - 1$ near $\xi = 1$, where $p \to 0$.

A spectral discretization of the radial equation (e.g., a mapped Chebyshev method with coordinate transformation $\xi = 1+t^2$ to resolve the $\xi = 1$ endpoint) would eliminate these issues and achieve exponential convergence. This is a straightforward extension left for future work.

Appendix B: Comparison with Failed Approaches

Before arriving at the separated solver, two other discretization strategies were attempted and found inadequate.

1. 1D axial lattice

The simplest possible molecular lattice places vertices along the internuclear axis at $z \in \{0, R/4, R/2, 3R/4, R\}$ with nearest- neighbor edges. This fails catastrophically because:

1. The 1D Coulomb potential $-Z/|z|$ is too singular without the r^2 Jacobian factor from 3D integration.
2. The grid spacing $h = R/4$ artificially couples the kinetic energy scale to R.
3. The 1D model has no physical correspondence to the 3D molecular problem.

A cylinder-averaged variant (averaging the Coulomb potential over a cylinder of radius ρ_0) produces qualitatively correct PES shapes but requires ρ_0 as a free parameter, achieving only $\sim 30\%$ energy error at best.

2. 2D tensor-product grid

A direct 2D discretization on an $N_\xi \times N_\eta$ tensor product grid in (ξ, η) space solves the *unseparated* problem $H\Psi = EM\Psi$ where $M = \mathrm{diag}(\xi^2 - \eta^2)$ is the volume weight. While correct in principle, this approach suffers from:

1. Slow convergence: 28% error at $50 \times 25 = 1250$ unknowns, versus 1.01% error at $N_\xi = 5000$ for the separated solver.
2. Boundary condition difficulties: Dirichlet BCs at $\eta = \pm1$ force $\Psi = 0$ at the nuclei, which is wrong for σ_g states. The correct natural zero-flux BCs are nontrivial to implement on a 2D grid.
3. Computational cost: the 2D generalized eigenvalue problem is $O(N^3)$ in the total number of grid points, versus $O(N_\xi)$ for the separated tridiagonal solver.

The separated approach is $\sim 50\times$ more efficient because it exploits the exact separability of the two-center Coulomb problem.

[1] J. Loutey, "The Geometric Atom: Quantum State Space as a Packing Problem," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Geometric Atom: Quantum Mechanics

as a Packing Problem," GeoVac Paper 1 (2026).
[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).
[4] J. Loutey, "Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis," GeoVac Paper 8–9 (2026).
[5] J. Loutey, "Topological LCAO Full CI for Heteronuclear Diatomics," GeoVac LCAO-FCI Paper (2026).
[6] J. Loutey, "Full configuration interaction on a sparse graph Hamiltonian: Multi-electron atoms via relational lattice indexing," GeoVac FCI Paper (2026).
[7] J. Loutey, "The Nuclear Lattice: Discrete Graph Structures for Molecular Vibration and Rotation," GeoVac Paper 10 (2026).
[8] V. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145–154 (1935).
[9] D. R. Bates, K. Ledsham, and A. L. Stewart, "Wave functions of the hydrogen molecular ion," Philos. Trans. R. Soc. London A **246**, 215–240 (1953).
[10] P. M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw-Hill, New York, 1953).
[11] C. Flammer, *Spheroidal Wave Functions* (Stanford University Press, Stanford, 1957).
[12] H. M. James and A. S. Coolidge, "The ground state of the hydrogen molecule," J. Chem. Phys. **1**, 825–835 (1933).
[13] R. P. Brent, *Algorithms for Minimization without Derivatives* (Prentice-Hall, Englewood Cliffs, NJ, 1973).
[14] L. N. Trefethen, *Spectral Methods in MATLAB* (SIAM, Philadelphia, 2000).